

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 52 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ohm's Law, Drift Velocity & Resistance Temperature	29 PYQs	Conduction
Very High	Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	24 PYQs	DC Instruments

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ohm's Law, Drift Velocity & Resistance Temperature	$I = n e A v_d$, $v_d = \frac{e E \tau}{m}$, $\rho = \frac{m}{n e^2 \tau}$, $R(T) = R_0(1 + \alpha \Delta T)$, $\vec{J} = \sigma \vec{E}$
Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	$\frac{P}{Q} = \frac{R}{S}$ (Wheatstone), $\frac{E_1}{E_2} = \frac{I_1}{I_2}$, $r = R \left(\frac{I_1}{I_2} - 1 \right)$, $S = \frac{I_g G}{I - I_g}$, $R_{\text{series}} = \frac{V}{I_g} - G$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ohm's Law, Drift Velocity & Resistance Temperature (Very High Yield — 29 PYQs)

Core Principle: Microscopic model of electrical conduction, electron mobility $\mu = v_d/E$, and thermal variation of resistivity.

■■ **Caution / JEE Pitfall:** For conductors, $\alpha > 0$ (resistance increases with temperature). For semiconductors, $\alpha < 0$ (resistance decreases with temperature).

■ **Speed Trick:** When a wire is stretched to n times its initial length while preserving volume, its resistance becomes $n^2 R$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two resistors of $200\ \Omega$ and $400\ \Omega$ are connected in series with a battery of 100 V . A bulb rated at 200 V , 100 W is connected ac..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "An electric power line having total resistance of $2\ \Omega$, delivers 1 kW of power at 250 V . The percentage efficiency of transmission li..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "A battery with EMF E and internal resistance r is connected across a resistance R . The power consumption in R will be maximum when:..." [Ans: D]

3.2 Kirchhoff's Laws, Wheatstone Bridge & Potentiometer (Very High Yield — 24 PYQs)

Core Principle: Kirchhoff's junction and loop rules, balanced bridges, potentiometer EMF comparison, and galvanometer conversion.

■ **Caution / JEE Pitfall:** A potentiometer measures EMF with zero drawn current, making it infinitely superior to a real voltmeter which always draws finite current.

■ **Speed Trick:** To convert galvanometer to ammeter: connect small shunt S in parallel. To voltmeter: connect high resistance R in series.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Refer to the figure given below. The values of I_1, I_2 and I_3 are ____" [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "Two resistors of 100Ω each are connected in series with a 9 V battery. A voltmeter of 400Ω resistance is connected to measure ..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Refer to the figure given below. The values of I_1, I_2 and I_3 are ____ .

- (A) $I_1=2.5A, I_2=1.875A, I_3=1.875A$ [Correct]
- (B) $I_1=1.875A, I_2=2.5A, I_3=1.875A$
- (C) $I_1=1.875A, I_2=1.875A, I_3=2.5A$
- (D) $I_1=2.5A, I_2=2.5A, I_3=1.875A$

Correct Answer: A

Solution Analysis: Using Kirchhoff's current law, the net accumulation of current at a junction is zero. So, at junction A, $I_1 - I_3 - I_{AD} = 0 \Rightarrow I_{AD} = I_1 - I_3$ And at junction D, $I_{AD} + I_2 - I_{DB} = 0 \Rightarrow I_1 - I_3 + I_2 - I_{DB} = 0 \Rightarrow I_{DB} = I_1 - I_3 + I_2$ Applying Kirchhoff's voltage law in closed loop I, $-4(I_1 - I_3) - 2(I_1 + I_2 - I_3) - I_1 + 10 = 0$
 $-7I_1 - 2I_2 + 6I_3 = -10$... (i) In loop (II) $I_1 + 10 - 4I_3 - 2(I_3 - I_2) = 0$ $2I_2 - 6I_3 - I_1 = -10$... (ii) In loop III, $-5 - 2(I_3 - I_2) + 2(I_1 + I_2 - I_3) = 0$ $2I_1 + 4I_2 - 4I_3 = 5$... (iii) From e

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#1)

Two resistors of 200Ω and 400Ω are connected in series with a battery of 100 V. A bulb rated at 200 V, 100 W is connected across the 400Ω resistance. The potential drop across the bulb is ____ V.

- (A) 25
- (B) 50 [Correct]
- (C) 66.6
- (D) 100

Correct Answer: B

Solution Analysis: The bulb is rated at power $P=100W$ and the voltage rating is $V_{(rated)}=200V$. Using the power formula $P=\frac{V^2}{R}$, the resistance of the bulb is, $R_{(b)}=\frac{V_{(rated)}^2}{P} \Rightarrow R_{(b)}=\frac{(200)^2}{100}=(40000/100)=400\Omega$ The bulb (400Ω) is connected in parallel with the 400Ω resistor. The equivalent resistance ($R_{(p)}$) of two equal resistances in parallel is : $\frac{1}{R_{(p)}}=(1/400)+(1/400)=(2/400) \Rightarrow R_{(p)}=200\Omega$ This parallel combination is in series with the 200Ω resistor. $R_{(t)}$